Hyewon Jung

hyewonj@andrew.cmu.edu

EDUCATION

Carnegie Mellon University, College of Engineering Pittsburgh, PA
Master of Science in Electrical and Computer Engineering 08/2026 - 12/2027

- Current Relevant Courses: Foundations of Computer Systems

Johns Hopkins University, Whiting School of Engineering Baltimore, MD
Bachelor of Science in Electrical Engineering, Computer Science | GPA: 3.91/4.0 08/2022 - 05/2026

- Dean's List Fall 2022 - Spring 2026
- Completed Relevant Courses: Intermediate Programming, Computer Systems Fundamentals, Mastering Electronics I/II, Computer Vision, Deep Learning, Intro Algorithms, Artificial Intelligence, Capstone Design Project I/II

SKILLS

Programming Languages: Python, C, C++, Java
Frameworks and Libraries: scikit-learn, PyTorch, Numpy, Pandas, Matplotlib, OpenCV
Tools: Git, Visual Studio, Vivado, IntelliJ IDEA, MATLAB, Eclipse, Google Colab, Cursor

RESEARCH AND PROJECTS

ARCADE Lab at Johns Hopkins University - Explainable AI (XAI) for Surgical Feedback Generation Baltimore, MD
Undergraduate Research Assistant 10/2025 - 06/2026

- Developed computer vision pipelines for surgical skill assessment, including data preprocessing and 2D-to-3D extension of movement analysis for explainable feedback generation.
- Implemented Dynamic Time Warping (DTW) algorithms to enable temporal alignment of trainee trajectories against expert benchmarks to provide time-specific and actionable feedback to surgeons-in-training.

Senior Capstone Design - Assistive Board Game Device Project Baltimore, MD
Project Lead, Lead Engineer 01/2026 - 05/2026

- Led end-to-end development of a CV-based assistive, multisensory feedback device utilizing computer vision and haptics to enable children with visual impairments to play board games fairly with sighted peers.
- Implemented a real-time object detection pipeline using YOLOv26n to track game state, translating visual data into non-visual cues via engineered tactile, audio, and haptic feedback mechanisms.

Bai Lab at Johns Hopkins Bayview Medical Center - Cardiac MR Sequence Classification Baltimore, MD
Undergraduate Research Assistant 09/2024 - 08/2025

- Explored adaptation of an unsupervised clustering algorithm from brain MRI sequence classification to cardiac MRI, including data preprocessing and preliminary evaluations. Paper accepted to the Machine Learning for Clinical Decision Support (ML-CDS) Workshop at MICCAI 2025.

WORK EXPERIENCE

JHU Electrical & Computer Engineering Dept. Baltimore, MD
Lab Assistant 01/2024 - 02/2026

- Performed various tasks including repairing, re-conditioning, and calibrating lab equipment for undergraduate/graduate lab courses, documenting new purchases, organizing stockroom, and maintaining electronics facilities across JHU Homewood campus.
- Assisted the Lab Coordinator during office hours for Gateway: Python and Mastering Electronics I/II courses to answer students' questions and help debug code and circuits.

Korea Innovation Center Washington DC (KIC DC) Berkeley, CA
Tech Frontier Program Technical Translator Intern 05/2025 - 07/2025

- Facilitated communication between 28 Korean startup teams and business mentors at UC Berkeley, Haas School of Business, by providing translation based on both technical and business knowledge.